

# DevOps Fundamentals: Study Notes

## 1. What is DevOps?

DevOps is a set of practices that brings software development and IT operations together. The goal of DevOps is to shorten the time between writing code and running it safely in production. It relies on automation, shared ownership and fast feedback loops.

Before DevOps, developers and operations teams often worked in separate silos. Developers wanted to ship new features quickly, while operations teams wanted stability. DevOps reduces this conflict by making both teams responsible for the full lifecycle of the software.

## 2. Version Control with Git

Git is a distributed version control system created by Linus Torvalds in 2005. Every developer has a full copy of the repository history on their own machine. A commit is a snapshot of the project at a point in time, and every commit is identified by a SHA-1 hash.

A branch is a movable pointer to a commit. Teams usually keep a stable main branch and create feature branches for new work. A pull request is a request to merge one branch into another, and it is the usual place where code review happens.

## 3. Continuous Integration

Continuous Integration (CI) is the practice of merging code changes into a shared branch frequently, often several times a day. Every merge triggers an automated build and an automated test run. If a test fails, the team is notified immediately and fixes the build before adding new work.

The main benefit of CI is that integration problems are found early, when they are small and cheap to fix. A CI server such as Jenkins watches the repository and runs the pipeline on every change.

## 4. Continuous Delivery and Continuous Deployment

Continuous Delivery means that every change which passes the pipeline is ready to be released, but a human still decides when to release it. Continuous Deployment goes one step further: every change that passes all stages is released to production automatically, with no manual approval.

A deployment pipeline is a sequence of stages such as build, test, scan, package and deploy. A failure in any stage stops the pipeline, so broken code never reaches

users.

## 5. Jenkins

Jenkins is an open-source automation server written in Java. A Jenkins pipeline can be defined in a text file called a Jenkinsfile, which is stored in the same repository as the application code. This idea is called pipeline as code.

A declarative Jenkinsfile is made of stages, and each stage contains steps. The post section of a pipeline runs after the stages finish and can react to success or failure, for example by sending a notification or rolling back a deployment. Jenkins stores secrets such as passwords and tokens in its credentials store, and masks them in build logs.

## 6. Containers and Docker

A container packages an application together with its libraries and configuration so that it runs the same way on every machine. Containers share the host operating system kernel, which makes them much lighter than virtual machines.

A Docker image is a read-only template made of layers. A container is a running instance of an image. A Dockerfile is a text file with instructions that Docker follows to build an image. Each instruction creates a new layer, and Docker reuses cached layers when their inputs have not changed.

A multi-stage build uses several FROM instructions in one Dockerfile. Only the final stage is shipped, so build tools and test dependencies do not end up in the production image. This keeps the image smaller and reduces its attack surface.

Docker Compose is a tool for defining and running multi-container applications using a YAML file called docker-compose.yml. A container registry such as Docker Hub stores and distributes images.

## 7. Image Tagging and Rollback

Every image can have one or more tags. The tag latest is only a convention and it keeps moving to the newest image. Tagging each image with the git commit SHA makes every build traceable to the exact code that produced it.

A rollback is the process of returning a system to a previous known-good version. When images are tagged by commit SHA, a rollback simply means deploying the previous SHA again.

## 8. Security Scanning

Trivy is an open-source vulnerability scanner made by Aqua Security. It compares the operating system packages and language libraries inside an image against databases of known vulnerabilities. Each known vulnerability has an identifier called a CVE, which stands for Common Vulnerabilities and Exposures.

Vulnerabilities are graded by severity: low, medium, high and critical. Many pipelines fail the build only on high and critical findings that already have a fix available. Running containers as a non-root user is another simple practice that limits the damage an attacker can do.

## 9. Testing and Quality Gates

Unit tests check a single function in isolation. Integration tests check that several components work together correctly. Code coverage measures the percentage of code lines that are executed by the tests. A quality gate is a rule in the pipeline that fails the build when a metric falls below a threshold, for example when coverage drops below 80 percent.

## 10. Monitoring and Observability

Monitoring tells a team whether a system is healthy. Prometheus is an open-source monitoring system that collects metrics by scraping HTTP endpoints at regular intervals. Grafana is commonly used to build dashboards on top of Prometheus data. A health check is a small endpoint or command that reports whether a service is able to handle requests.